University of Central Punjab
Faculty of Information Technology
Midterm lab exam– Spring 2026

Subject	Software Construction and Development	Course Code	SESD3241
Sections			
Allowed Time	90 minutes	% contribution towards final lab evaluation	

INSTRUCTIONS

1. Read the tasks carefully.
2. All questions are mandatory. Only appropriate queries would be entertained by the lab instructor.
3. Exchange of solution is not allowed.
4. Submit the exam solution within prescribed time.

Question 1:

A parking system needs a program to calculate the **parking fee** of a vehicle based on the number of hours parked.

Write a Java program that:

- Takes input for **number of hours parked**
- Calculates the total parking fee based on the following rules:

Parking Charges:

- First **2 hours** → Rs. 100 per hour
- Next **3 hours (3–5 hours)** → Rs. 80 per hour
- Beyond **5 hours** → Rs. 50 per hour

Display:

- Total hours parked
- Total parking fee

Question 2:

You are given the following Java code for a **delivery cost calculation system**. The system currently supports Standard and Express delivery. The company plans to add more delivery types (e.g., Same Day, International) in the future.

Task:

1. Identify which **SOLID principle is violated**. Explain why it is a violation.
2. Now, the system needs to support **Same Day Delivery** with cost = weight * 150. Update the design so future delivery types can be added easily.
3. Refactor and rewrite the code so that new delivery types can be added **without modifying existing code**.

```java
public class DeliveryService {

    public void calculateCost(String type, int weight) {

        if (type.equals("Standard")) {
            System.out.println("Cost: " + (weight * 50));
        }
        else if (type.equals("Express")) {
            System.out.println("Cost: " + (weight * 100));
        }
        else {
            System.out.println("Invalid Delivery Type");
        }
    }

    public static void main(String[] args) {
        DeliveryService ds = new DeliveryService();
        ds.calculateCost("Standard", 5);
    }
}
```

Question 3:

You are given the following Java code for a **movie ticket booking system**. The code is functional but contains several design issues.

Task:

1. Identify **at least three code smells**
2. Explain each smell
3. Refactor and rewrite the improved code

```
public class ticketSystem {

    public void calcPrice(String type, int seats) {

        if (type.equals("normal")) {
            int price = seats * 300;
            if (seats > 5) {
                price = price - 100;
            }
```

```
            System.out.println("Price: " + price);
        }
        else if (type.equals("vip")) {
            int price = seats * 600;
            if (seats > 5) {
                price = price - 200;
            }
            System.out.println("Price: " + price);
        }
        else {
            System.out.println("Invalid Type");
        }
    }

    public void CalcTax(String type, int seats) {

        if (type.equals("normal")) {
            int tax = seats * 20;
            System.out.println("Tax: " + tax);
        }
        else if (type.equals("vip")) {
            int tax = seats * 40;
            System.out.println("Tax: " + tax);
        }
    }

    public static void main(String[] args) {
        ticketSystem ts = new ticketSystem();
        ts.calcPrice("normal", 6);
        ts.CalcTax("normal", 6);
    }
}
```